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CHAPTER

GAUSS'S LAW

Flat surface and $\hat{n} = 90^\circ$ on plane means $\perp$

Q.1. Define and explain electric flux?

ELECTRIC FLUX

The total number of electric lines of force passing through certain area is called electric flux. It is denoted by ϕ_e .

The scalar product of electric field $\vec{E}$ and vector area $d\vec{a}$ of a surface S is equal to electric flux.

$$d\phi_e = \vec{E} \cdot d\vec{a}$$

The net electric flux through entire surface is

$$\phi_e = \oint \vec{E} \cdot d\vec{a}$$

Here $\vec{E}$ is electric field and $d\vec{a}$ is area vector.

Where $d\vec{a}$ is called vector area whose magnitude is the area of the surface element and direction is outward drawn normal to surface. The circle on integral means closed surface.

The SI unit of electric flux is called $\text{N}\cdot\text{m}^2/\text{C}$ MAXIMUM FLUX

The electric flux through a surface will be maximum when surface is held perpendicular to electric field lines. The angle between normal $\hat{n}$ and electric field lines $\vec{E}$ is zero.

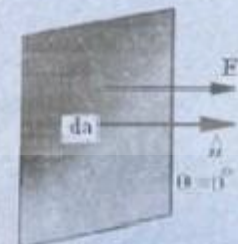
if there is a part or portion of area use d with vector area.

$$\phi_e = \int \vec{E} \cdot d\vec{a}$$

$$\phi_e = \int E da \cos\theta$$

$$\phi_e = \int E da \cos 0^\circ = \int E da$$

Unit vector $\times$ anything = vector.



MINIMUM FLUX

When surface is held parallel to electric field lines, the angle between normal and electric field lines is 90° . In this case, electric flux through surface is minimum.

derivation
and
Integration

$$\phi_e = \int \vec{E} \cdot d\vec{a}$$

$$\phi_e = \int E da \cos\theta$$

$$\phi_e = \int E da \cos 90^\circ = 0$$



Q.2 State and prove Gauss's law and write its integral and differential form.

GAUSS'S LAW We use to find electric field intensity.

The Gauss's law states that electric flux passing through any closed surface is equal to $\frac{1}{\epsilon_0}$ times charge enclosed by that surface.

PROOF of Gauss's law $\phi_e = \frac{q}{\epsilon_0}$ total, net

Consider a +q charge is placed in a closed surface. The electric field lines come out of the surface. Take small area element $da = da \hat{n}$ of the surface at point P. The normal to area element makes angle θ with electric field line. The electric field at point P due to +q charge having distance r is given as

$$\vec{E} = \frac{kq}{r^2} \hat{r}$$

The electric flux through area element da is

$$d\phi_e = \vec{E} \cdot d\vec{a}$$

The net flux through entire surface is

$$\phi_e = \int \vec{E} \cdot d\vec{a}$$

$$\phi_e = \int \frac{kq}{r^2} \hat{r} \cdot da \hat{n}$$

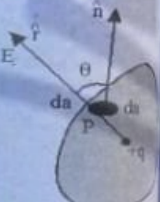
$$\phi_e = kq \int \frac{da}{r^2} \hat{r} \cdot \hat{n} = kq \int d\Omega = 4\pi$$

Where $\int d\Omega$ is called total solid angle and its value is 4π

$$\phi_e = \frac{q(4\pi)}{4\pi\epsilon_0}$$

$$\phi_e = \frac{q}{\epsilon_0}$$

Hence, Gauss's law is proved.



$$\int \left(\frac{da}{r^2} \right) = \Omega$$

CHAPTER 3) GAUSS'S LAW

INTEGRAL FORM

The volume charge density is defined as

$$\rho = \frac{dq}{dV}$$

Here integrate on b.s.w $\frac{dq}{dV} = \rho$
 $q = \int \rho dV$ (1)

The net flux through entire surface is

$$\phi_e = \int \vec{E} \cdot d\vec{a}$$

Put eq (1) and eq (2) in Gauss's law

$$\phi_e = \frac{q}{\epsilon_0}$$

$$\int \vec{E} \cdot d\vec{a} = \frac{1}{\epsilon_0} \int \rho dV$$

Mostly used
in numericals

This is called integral form of Gauss's law

DIFFERENTIAL FORM

We know that

$$\int \vec{E} \cdot d\vec{a} = \frac{1}{\epsilon_0} \int \rho dV$$

Apply divergence theorem changes surface integral into volume integral

$$\int \vec{E} \cdot d\vec{a} = \int (\vec{\nabla} \cdot \vec{E}) dV$$

Compare eq (1) and eq (2)

$$\int (\vec{\nabla} \cdot \vec{E}) dV = \frac{1}{\epsilon_0} \int \rho dV$$

$$\int (\vec{\nabla} \cdot \vec{E}) dV = \frac{1}{\epsilon_0} \int \rho dV$$

$$\int (\vec{\nabla} \cdot \vec{E} - \frac{\rho}{\epsilon_0}) dV = 0$$

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

This is called differential form of Gauss's law

Define or state Gauss law and write its integral and differential form

State for electric field and derive Coulomb's Gauss's law? Using integral form of Gauss's law derive a find intensity Of electric field at a point due

EdaCos0
0

LAW causs's law states that electric pas*ing Closed to — times Charge by that surface,

GAUSS'S LA w

DEDUCTION OF cot LAW FROM GAUSS'S LAW
•q charge pt.-«cd at the cenier of having radius called gaussi;m surface. rhe electric lines of are outward perpendicular

eq(3' and

Take da Of sphere. The unit vector n indicat&s the Of da and perpendicular to surface angle between n and E zero. The through surface Of spin: re is

is statement of Coulomb's law is deduced from

Calculate electric field due to innnite line charge by using Gauss's

= f

Edacose

f

FIELD TO LINE OV CHARGE

a Straight wire having infinite negligible
Now lake a cylinder Surface having r height b which
part Of Wire in fig.

The P is the on the or the
Take a small cylinder where want field,
length
element of the line having dragedq. The linear charge
densil' is given as E

The Gauss's law States that electric flux passing through surface Ofspbereis

to — charge enclosed by the surface of sphere. net charge enclosed by the cylinder height h is

and cq(2)

The cylinder has three surfaces

O Top circular surface

O Bottom circular surface

O Cur:cd Su dace

The elCQtric field lines will bc along radius Of cylinder. Take a small area
element da Of each surface of lhe cylinder. The angle tRtween normal u to elenp:m
oceuc•.cd sgrfRc and electric field lines is tem.

Tlle electric force per unit charge is called electric field when a charge i." circuaRr surface and field lines is90'.

The angle iwteen normal n element top circular surface and bottom in Illis field.

Electric through top surface is

NITE SHEET OF CHARGE

having infinite length and non-conducting

- jEdaCos0 Wc to calculate electric field Eat points near the shret by using Gauss's , - JEdaCos900
Electric flux through circular Sur race is cylinder cylinder through called gaussian Take Small surrace having da of Sheet having area A. Nowdq.

surfJCC charge density is defined as

0: JEdaCos900

Electric

curved surface is Area (hr) h charge Of Shecl
enclosed by A Ofeylindet is

E da Cosb

JEdaCos00 - JEda
E'2rrh)

The net flux through entire surface of cylinder is.

The ga"SSan cylinder has

O Top capped surface

O Bottom capped surface

O Curved surface

O E(2arh) The electric field E is parallel to end caps*d surfaces. The are at equal distance from sheet. field E has same at

Put

and eq(2) in Gauss's law The cylilbder has
three surfaces so flux through each surface is
given

Electric flux through top-caped surface is

JE .da

- jEdaCose

In vector form

Electric flux through bottom-capped surface is

— $\oint \mathbf{E} \cdot d\mathbf{a}$ an

$$\oint \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} \sum q_i$$

Q.5 State Gauss's law and apply it to calculate electric field near a uniformly charged sheet of charge?

Electric flux through curved surface is

GAUSS'S LAW

The Gauss's law states that electric flux passing through any closed surface is $\frac{1}{\epsilon_0}$ times charge enclosed by that surface.

Th. entire of cylinder is

The volume charge density is defined as

$\rho = \frac{q}{V}$

$$E = \frac{\sigma}{\epsilon_0}$$

in Gauss's law is.

$$\oint \mathbf{E} \cdot d\mathbf{a}$$

point p at

center of charged sphere where we want

electric and Imagine charge is distributed uniformly in a sphere of radius R. P lies on the surface of the sphere. The angle between the radial field E and normal n to area da is

vector form

through

is

$$\oint \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} \sum q_i$$

net gaussian sphere is

Q.6 State Gauss's law and apply it to calculate distribution of electric

field intensity point due to spherical volume charge

I in Gauss's

GAUSS'S LAW states electric passing through any closed surface

to $\frac{1}{\epsilon_0}$ times surface.

ELECTRIC FIELD DUE TO SPHERICAL CHARGE DISTRIBUTION

The electric field due to spherical volume charge distribution using Gauss's law is divided into two parts

- Electric field inside spherical charge distribution
- Electric field outside spherical volume charge distribution
- Electric field on surface of spherical volume charge distribution

ELECTRIC FIELD OUTSIDE SPHERICAL CHARGE DISTRIBUTION

Consider a sphere of radius R having positive charge. Take a small volume element dV of the sphere having charge dq.

The field in vector outside of point p having r center of sphere is

ELECTRIC FIELD INSIDE SPHERICAL CHARGE DISTRIBUTION

Consider a uniformly charged sphere having radius R. Take a small volume element dV of sphere having charge dq. The charge density is defined as

$$dq = \rho dV$$

The net charge enclosed in sphere is

$$dq = \rho dV$$

The charge enclosed in sphere is

Now take a point p inside charged sphere having distance r from center of sphere where we want to calculate electric field. Imagine a Gaussian sphere of radius r such that p lies on its surface.

Charge enclosed in Gaussian sphere of radius r is

Divide eq (2) and

$$q = \rho \frac{4}{3}\pi r^3$$

consider small area element da of Gaussian sphere. The angle

electric field E is normal n to area element da is through area element da is

$$\oint E \cdot da \cos\theta = \oint E da \cos\theta$$

The net flux is

$$E(4\pi r^2)$$

Put eq (3) and eq (4) in Gauss's law

The electric field inside charge sphere is

Put R to calculate electric field at the surface of sphere

Gauss's law and apply it to calculate electric field intensity due to charged shell of constant surface charge density.

GAUSS'S LAW states that electric flux passing through any closed surface is equal

to the times charge enclosed by that surface.

SHELL THEOREMS are two shell theorems derived by using uniform

charge density shell of charge having constant surface charge density is stated as

1st SHELL THEOREM

A uniform spherical shell of charge center behaves for external points in such a way that all its charge was concentrated

PROOF

Consider a uniformly charged spherical shell such as charge sprayed uniformly over the surface of a balloon having radius R. Take a small da of

charge

The surface charge density σ is defined as

$$\sigma = \frac{dq}{da}$$

$$dq = \sigma da$$

The net charge on the shell

Consider a point P outside shell having distance r from the center of spherical charged shell where we want to evaluate electric field. Imagine a Gaussian spherical shell which encloses charged shell such that P lies on its surface

/ ELECTRIC FIELD AT SURFACE OF

SPHERE

Now take a small area element da of shell having radius r . The angle between normal n to this area element and electric field E is zero. Electric flux through area element da is

The net flux is

Put eq (2) in Gauss's law

- The net flux is

Put eq (2) in Gauss's law.

Inside the Gaussian surface, charge is zero ($q = 0$)

The electric force on charge placed inside gaussian shell.

It means a uniform charged spherical shell exerts no electrostatic force on a charge placed inside the shell. Hence, second shell theorem is proved.

by using Gauss's law that an excess charge placed on a conductor moves entirely to the outer surface of the conductor and none of the excess charge is found within the body of the conductor. So, the electric field inside the conductor is zero.

This is electric field at point p due to charge q at its center. It can be shown that the surface of a hollow spherical shell is concentrated at its center.

SECOND SHELL THEOREM

A uniform charged spherical shell exerts no electrostatic force on a charge placed inside the shell.

PROOF

Consider a positively charged hollow spherical shell of radius R and uniform charge density σ . The net electric field at a point p inside the shell is to be found.

Take a point p inside the shell where we want to evaluate electric field. Let its distance from the center of the spherical charged shell be r . Imagine a Gaussian surface which encloses no charge such that p lies on its surface.

Now consider a small area element da on the Gaussian shell having radius r . The

angle between the normal to the area element da and the electric field E is θ . The electric flux through the area element da is $E da \cos \theta$.

The total electric flux through the Gaussian surface is

Consider a hollow spherical conductor of radius R which is shown as a Gaussian surface. The electric field inside the conductor is to be found.

Since the conductor is in electrostatic equilibrium, the electric field inside the conductor is zero. If a Gaussian surface is drawn inside the conductor, the net electric flux through it is zero. Hence, the net charge enclosed by the Gaussian surface is zero. This means that all the excess charge is on the outer surface of the conductor.

If a Gaussian surface is drawn inside the conductor, the net electric flux through it is zero. The value of electric field inside the conductor is zero.

means by Gauss's law electric flux through Gaussian surface which is inside charged conductor must be zero.

means net charge $q = 0$, If the Charge is not inside the gaussian surface. it be outside the surface The surf-acc is very close to outer surface of the as in fig. Hence. charge Q added is on outer surface of the conductor.

is concluded that an excess charge placed on an isolated conductor relies to the outer surface of the conductor. Of the, excess charge is found on the body conductor

0.9 Calculate electric field outside the charged isolated conductor by using Gauss's law?

EI-EC-IRIC FIELD

Consider a positively charged isolated conductor, excess charge placed on a lated conductor moves to the outer surface of the conductor. In a gaussian surface in the shape of cylinder.

The cylinder has three portions, The end caps are parallel to the surface of the cylinder and One end is inside and other is outside the surface as shown in fig. The walls of cylinder are perpendicular to surface The end capped surface The electric field just a charged isolated conductor surface of conductor. The value of Electric through end capped outside

If dq charge is enclosed area of gaussian surface. The surface charge is defined as

Let q - Total net charge enclosed by end which is outside having area A is

The Gauss's Law

form

This electric field

outside the charged

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\epsilon_0}$$

The field inside a conductor is Zero. through end cap inside surface is

$$\vec{E} \cdot d\vec{A} = E dA \cos \theta$$

angle θ is
The Q_{enc} , or through

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\epsilon_0} \cos \theta$$

The flux is

SHORT QUESTIONS

Suppose that an to be decreasing field in in some Strength region in that is direction. to What constant do you conclude about the charge in the region?

The electric field and charge are related as $E \propto r$. If that strength of electric field decreases When Strength of decreases. Of remains constant

IS it precisely true that Gauss's law States that the of lines of crossing any closed surface in the outward to the net positive charge enclosed the The law States that total electric flux is to enclosed by that surface.

Therefore, it is precisely true that Gauss's law states that the total of of flux crossing any closed surface in the Outward direction called electric flux is proportional to the net positive within the

A point charge is placed at the center of a spherical gaussian surface. Is flux Changed (a) if the surface replaced by cube the (b) If the sphere is replaced

by a cube of tenth the volume (C) If the is moved Off center in the original sphere.
Still remaining If second inside Charge (d) is If the charre andis moved just
outside the origin•l sphere outside the original sphereond (O If a second charge
is placed inside
gaussian surface.

If
s: A point charge is placed at the center of a spherical gaussian surface is
replaced by a cube of same volume. the will not 'hange,

A B)int charge is placed at the center Of a s/hctical gaussian surface. If the sphere
is replaced by a cube of otwtenth the volume, the flux will not change.

